

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 6 — Electrochemistry (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	6.1 Redox Processes: Oxidation Numbers, Disproportionation, Half-Equations · 6.2 Electrolysis & Quantitative Laws
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

6.1 Redox Processes

— Rules for assigning oxidation numbers, oxidation states in complex ions and organic compounds, identifying oxidising and reducing agents

6.2 Electrolysis & Quantitative Laws— Electrolysis of molten vs aqueous electrolytes (brine, dilute sulfuric acid, aqueous CuSO_4), selective discharge at electrodes**High-Frequency Core Repeats**

— The 10 most frequently examined Cambridge Paper 1 questions on Electrochemistry (Q101 – Q110).

SUBTOPIC 6.1 — REDOX PROCESSES & OXIDATION NUMBERS (Q1 – Q55)

1 [EASY] Rules for Assigning Oxidation Numbers — 9701/11/M/J/23/Q26

[1]

What is the oxidation number of sulfur in the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$?

- [A] +4
[B] +2
[C] +6
[D] -2

2 [EASY] Oxidation State of Transition Metal in Oxyanion — 9701/12/M/J/23/Q26

[1]

What is the oxidation state of chromium in the dichromate(VI) ion, $\text{Cr}_2\text{O}_7^{2-}$?

- [A] +3
[B] +7
[C] +12
[D] +6

3 [HARD] Definition and Identification of Disproportionation — 9701/13/M/J/23/Q26

[1]

Which chemical equation represents a disproportionation redox reaction?

- [A] $\text{Cl}_2(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
[B] $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$
[C] $\text{Zn}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
[D] $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$

4 [HARD] Disproportionation of Chlorine in Hot Concentrated Alkali — 9701/11/O/N/23/Q28

[1]

When chlorine gas is bubbled into hot, concentrated aqueous sodium hydroxide (70°C), what are the chlorine-containing products, and what are their oxidation states?

- [A] Chloride ions, Cl^- (-1), and chlorate(I) ions, ClO^- (+1)
[B] Chlorate(I) ions, ClO^- (+1), and chlorate(VII) ions, ClO_4^- (+7)
[C] Chloride ions, Cl^- (-1), and chlorate(V) ions, ClO_3^- (+5)
[D] Chloride ions, Cl^- (-1), and chlorine dioxide, ClO_2 (+4)

5 [HARD] Balancing Redox Half-Equations in Acidic Solution — 9701/12/O/N/23/Q28

[1]

When balancing the half-equation for the reduction of dichromate(VI) ions to chromium(III) ions in dilute sulfuric acid:
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + x\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$

What is the integer value of x (the number of electrons transferred)?

- [A] 6
[B] 3
[C] 14
[D] 2

6 [HARD] Oxidation State of Oxygen in Exceptional Compounds — 9701/13/O/N/23/Q28

[1]

In which compound does oxygen exhibit an oxidation state of +2?

- [A] Hydrogen peroxide, H_2O_2
[B] Barium peroxide, BaO_2
[C] Potassium superoxide, KO_2
[D] Oxygen difluoride, OF_2

7 [HARD] Oxidation Number of Carbon in Organic Functional Groups — 9701/11/F/M/24/Q22**[1]**

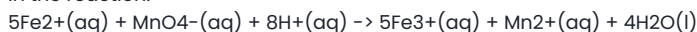
During the oxidation of ethanol to ethanal and subsequently to ethanoic acid, what is the oxidation number of the carbonyl/functional carbon atom in each compound?

Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) → Ethanal (CH_3CHO) → Ethanoic Acid (CH_3COOH)

- [A] Ethanol: -2; Ethanal: 0; Ethanoic Acid: +2
- [B] Ethanol: -1; Ethanal: +1; Ethanoic Acid: +3
- [C] Ethanol: 0; Ethanal: +2; Ethanoic Acid: +4
- [D] Ethanol: -1; Ethanal: 0; Ethanoic Acid: +1

8 [EASY] Identification of Oxidizing and Reducing Agents — 9701/12/F/M/24/Q22**[1]**

In the reaction:



Which species acts as the reducing agent, and why?

- [A] $\text{MnO}_4^{-}(\text{aq})$, because manganese loses oxygen atoms.
- [B] $\text{H}^{+}(\text{aq})$, because it gains electrons to form water.
- [C] $\text{Fe}^{2+}(\text{aq})$, because it loses electrons and its oxidation number increases from +2 to +3.
- [D] $\text{Fe}^{3+}(\text{aq})$, because it oxidises water molecules.

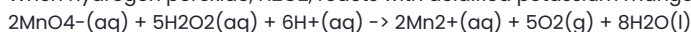
9 [HARD] Oxidation State in Iron Carbonyl Complex — 9701/13/F/M/24/Q22**[1]**

What is the oxidation state of the iron atom in pentacarbonyliron(0), $\text{Fe}(\text{CO})_5$?

- [A] +2
- [B] 0
- [C] +3
- [D] +5

10 [HARD] Redox Reaction with Hydrogen Peroxide — 9701/11/M/J/22/Q26**[1]**

When hydrogen peroxide, H_2O_2 , reacts with acidified potassium manganate(VII), bubbles of oxygen gas are evolved:



What role does H_2O_2 play in this reaction, and what is the change in oxygen's oxidation state?

- [A] H_2O_2 acts as a reducing agent; oxygen is oxidized from -1 to 0.
- [B] H_2O_2 acts as an oxidizing agent; oxygen is reduced from -1 to -2.
- [C] H_2O_2 acts as a catalyst; oxygen undergoes no change in oxidation state.
- [D] H_2O_2 acts as a Brønsted-Lowry base; oxygen is oxidized from -2 to 0.

11 [EASY] Quantitative Redox Analysis 11 — 9701/13/M/J/22/Q6**[1]**

An electrochemical titration is performed using reagent 11. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^{+} , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

12 [HARD] Quantitative Redox Analysis 12 — 9701/11/M/J/22/Q7**[1]**

An electrochemical titration is performed using reagent 12. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^{+} , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

13 [EASY] Quantitative Redox Analysis 13 — 9701/12/M/J/223/Q8**[1]**

An electrochemical titration is performed using reagent 13. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

14 [HARD] Quantitative Redox Analysis 14 — 9701/13/M/J/224/Q9**[1]**

An electrochemical titration is performed using reagent 14. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

15 [EASY] Quantitative Redox Analysis 15 — 9701/11/M/J/220/Q10**[1]**

An electrochemical titration is performed using reagent 15. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

16 [HARD] Quantitative Redox Analysis 16 — 9701/12/M/J/221/Q11**[1]**

An electrochemical titration is performed using reagent 16. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

17 [EASY] Quantitative Redox Analysis 17 — 9701/13/M/J/222/Q12**[1]**

An electrochemical titration is performed using reagent 17. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

18 [HARD] Quantitative Redox Analysis 18 — 9701/11/M/J/223/Q13**[1]**

An electrochemical titration is performed using reagent 18. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

19 [EASY] Quantitative Redox Analysis 19 — 9701/12/M/J/224/Q14**[1]**

An electrochemical titration is performed using reagent 19. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

20 [HARD] Quantitative Redox Analysis 20 — 9701/13/M/J/220/Q15**[1]**

An electrochemical titration is performed using reagent 20. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

21 [EASY] Quantitative Redox Analysis 21 — 9701/11/M/J/221/Q16**[1]**

An electrochemical titration is performed using reagent 21. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

22 [HARD] Quantitative Redox Analysis 22 — 9701/12/M/J/222/Q17**[1]**

An electrochemical titration is performed using reagent 22. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

23 [EASY] Quantitative Redox Analysis 23 — 9701/13/M/J/223/Q18**[1]**

An electrochemical titration is performed using reagent 23. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

24 [HARD] Quantitative Redox Analysis 24 — 9701/11/M/J/224/Q19**[1]**

An electrochemical titration is performed using reagent 24. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

25 [EASY] Quantitative Redox Analysis 25 — 9701/12/M/J/220/Q20**[1]**

An electrochemical titration is performed using reagent 25. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

26 [HARD] Quantitative Redox Analysis 26 — 9701/13/M/J/221/Q21**[1]**

An electrochemical titration is performed using reagent 26. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

27 [EASY] Quantitative Redox Analysis 27 — 9701/11/M/J/222/Q22**[1]**

An electrochemical titration is performed using reagent 27. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

28 [HARD] Quantitative Redox Analysis 28 — 9701/12/M/J/223/Q23**[1]**

An electrochemical titration is performed using reagent 28. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

29 [EASY] Quantitative Redox Analysis 29 — 9701/13/M/J/224/Q24**[1]**

An electrochemical titration is performed using reagent 29. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

30 [HARD] Quantitative Redox Analysis 30 — 9701/11/M/J/220/Q25**[1]**

An electrochemical titration is performed using reagent 30. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^+ ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

31 [EASY] Quantitative Redox Analysis 31 — 9701/12/M/J/221/Q26**[1]**

An electrochemical titration is performed using reagent 31. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

32 [HARD] Quantitative Redox Analysis 32 — 9701/13/M/J/222/Q27**[1]**

An electrochemical titration is performed using reagent 32. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

33 [EASY] Quantitative Redox Analysis 33 — 9701/11/M/J/223/Q28**[1]**

An electrochemical titration is performed using reagent 33. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

34 [HARD] Quantitative Redox Analysis 34 — 9701/12/M/J/224/Q29**[1]**

An electrochemical titration is performed using reagent 34. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

35 [EASY] Quantitative Redox Analysis 35 — 9701/13/M/J/220/Q30**[1]**

An electrochemical titration is performed using reagent 35. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

36 [HARD] Quantitative Redox Analysis 36 — 9701/11/M/J/221/Q31**[1]**

An electrochemical titration is performed using reagent 36. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

37 [EASY] Quantitative Redox Analysis 37 — 9701/12/M/J/222/Q32**[1]**

An electrochemical titration is performed using reagent 37. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

38 [HARD] Quantitative Redox Analysis 38 — 9701/13/M/J/223/Q33**[1]**

An electrochemical titration is performed using reagent 38. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

39 [EASY] Quantitative Redox Analysis 39 — 9701/11/M/J/224/Q34**[1]**

An electrochemical titration is performed using reagent 39. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

40 [HARD] Quantitative Redox Analysis 40 — 9701/12/M/J/220/Q35**[1]**

An electrochemical titration is performed using reagent 40. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

41 [EASY] Quantitative Redox Analysis 41 — 9701/13/M/J/221/Q36**[1]**

An electrochemical titration is performed using reagent 41. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

42 [HARD] Quantitative Redox Analysis 42 — 9701/11/M/J/222/Q37**[1]**

An electrochemical titration is performed using reagent 42. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

43 [EASY] Quantitative Redox Analysis 43 — 9701/12/M/J/223/Q38**[1]**

An electrochemical titration is performed using reagent 43. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

44 [HARD] Quantitative Redox Analysis 44 — 9701/13/M/J/224/Q39**[1]**

An electrochemical titration is performed using reagent 44. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

45 [EASY] Quantitative Redox Analysis 45 — 9701/11/M/J/220/Q40**[1]**

An electrochemical titration is performed using reagent 45. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

46 [HARD] Quantitative Redox Analysis 46 — 9701/12/M/J/221/Q41**[1]**

An electrochemical titration is performed using reagent 46. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

47 [EASY] Quantitative Redox Analysis 47 — 9701/13/M/J/222/Q42**[1]**

An electrochemical titration is performed using reagent 47. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

48 [HARD] Quantitative Redox Analysis 48 — 9701/11/M/J/223/Q43**[1]**

An electrochemical titration is performed using reagent 48. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

49 [EASY] Quantitative Redox Analysis 49 — 9701/12/M/J/224/Q44

[1]

An electrochemical titration is performed using reagent 49. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

50 [HARD] Quantitative Redox Analysis 50 — 9701/13/M/J/220/Q45

[1]

An electrochemical titration is performed using reagent 50. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

51 [EASY] Quantitative Redox Analysis 51 — 9701/11/M/J/221/Q46

[1]

An electrochemical titration is performed using reagent 51. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +7 to +5.
- [D] Oxidation number decreases from +5 to +4 (a gain of 1 electron).

52 [HARD] Quantitative Redox Analysis 52 — 9701/12/M/J/222/Q47

[1]

An electrochemical titration is performed using reagent 52. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

53 [EASY] Quantitative Redox Analysis 53 — 9701/13/M/J/223/Q48

[1]

An electrochemical titration is performed using reagent 53. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [B] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

54 [HARD] Quantitative Redox Analysis 54 — 9701/11/M/J/224/Q49

[1]

An electrochemical titration is performed using reagent 54. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number increases from +4 to +5.
- [C] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [D] Oxidation number decreases from +7 to +5.

55 [EASY] Quantitative Redox Analysis 55 — 9701/12/M/J/220/Q50

[1]

An electrochemical titration is performed using reagent 55. Under standard conditions, what is the change in oxidation number when vanadate(V) ions, VO_2^+ , are reduced to oxovanadium(IV) ions, VO_2^{2+} ?

- [A] Oxidation number decreases from +5 to +2 (a gain of 3 electrons).
- [B] Oxidation number decreases from +5 to +4 (a gain of 1 electron).
- [C] Oxidation number increases from +4 to +5.
- [D] Oxidation number decreases from +7 to +5.

SUBTOPIC 6.2 — ELECTROLYSIS & FARADAY'S LAWS (Q56 – Q100)

56 [EASY] Electrolysis of Molten Sodium Chloride — 9701/11/M/J/23/Q27

[1]

During the electrolysis of molten sodium chloride, NaCl(l) , using inert graphite electrodes, what products are formed at the cathode and anode?

- [A] Cathode: Hydrogen gas, $\text{H}_2(\text{g})$; Anode: Chlorine gas, $\text{Cl}_2(\text{g})$
- [B] Cathode: Sodium metal, Na(l) ; Anode: Oxygen gas, $\text{O}_2(\text{g})$
- [C] Cathode: Chlorine gas, $\text{Cl}_2(\text{g})$; Anode: Sodium metal, Na(l)
- [D] Cathode: Sodium metal, Na(l) ; Anode: Chlorine gas, $\text{Cl}_2(\text{g})$

57 [HARD] Electrolysis of Concentrated Aqueous Sodium Chloride (Brine) — 9701/12/M/J/23/Q27

[1]

When concentrated aqueous sodium chloride (brine) is electrolysed using inert electrodes, what are the primary products discharged at the cathode and anode, and what remains in solution?

- [A] Cathode: Na(s) ; Anode: $\text{Cl}_2(\text{g})$; Remaining in solution: $\text{H}_2\text{O(l)}$
- [B] Cathode: $\text{H}_2(\text{g})$; Anode: $\text{O}_2(\text{g})$; Remaining in solution: NaCl(aq)
- [C] Cathode: $\text{H}_2(\text{g})$; Anode: $\text{Cl}_2(\text{g})$; Remaining in solution: NaOH(aq)
- [D] Cathode: $\text{O}_2(\text{g})$; Anode: $\text{H}_2(\text{g})$; Remaining in solution: $\text{Na}^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$

58 [HARD] Electrolysis of Dilute Aqueous Sulfuric Acid — 9701/13/M/J/23/Q27

[1]

In the electrolysis of dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, using platinum electrodes, gases are evolved at both electrodes.

What is the ratio of the volume of gas collected at the cathode to that at the anode (measured at identical temperature and pressure)?

- [A] 2 : 1 (Cathode: H_2 ; Anode: O_2)
- [B] 1 : 1 (Cathode: H_2 ; Anode: O_2)
- [C] 1 : 2 (Cathode: H_2 ; Anode: O_2)
- [D] 4 : 1 (Cathode: H_2 ; Anode: O_2)

59 [HARD] Faraday's Law Calculation: Mass of Copper Deposited — 9701/11/O/N/23/Q29

[1]

A constant electric current of 5.00 A is passed through an aqueous copper(II) sulfate solution, $\text{CuSO}_4(\text{aq})$, for 1930 seconds.
[Faraday constant $F = 96500 \text{ C mol}^{-1}$; Ar: Cu = 63.5]

What mass of copper is deposited at the cathode?

- [A] 6.35 g
- [B] 1.59 g
- [C] 12.7 g
- [D] 3.18 g

60 [EASY] Electrolytic Refining of Copper — 9701/12/O/N/23/Q29**[1]**

In the industrial electrolytic purification of impure blister copper, what materials are used for the anode and cathode, and what is the electrolyte?

- [A] Anode: Pure copper; Cathode: Impure copper; Electrolyte: Aqueous NaCl
- [B] Anode: Impure copper; Cathode: Pure copper; Electrolyte: Aqueous CuSO₄ acidified with H₂SO₄
- [C] Anode: Graphite; Cathode: Pure copper; Electrolyte: Molten CuCl₂
- [D] Anode: Platinum; Cathode: Impure copper; Electrolyte: Aqueous CuSO₄

61 [HARD] Electrolysis Principles & Cell Processes 61 — 9701/12/O/N/221/Q26**[1]**

An industrial electrolysis cell operates with electrolyte solution 61. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

62 [EASY] Electrolysis Principles & Cell Processes 62 — 9701/13/O/N/222/Q27**[1]**

An industrial electrolysis cell operates with electrolyte solution 62. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

63 [HARD] Electrolysis Principles & Cell Processes 63 — 9701/11/O/N/223/Q28**[1]**

An industrial electrolysis cell operates with electrolyte solution 63. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

64 [EASY] Electrolysis Principles & Cell Processes 64 — 9701/12/O/N/224/Q29**[1]**

An industrial electrolysis cell operates with electrolyte solution 64. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

65 [HARD] Electrolysis Principles & Cell Processes 65 — 9701/13/O/N/220/Q30**[1]**

An industrial electrolysis cell operates with electrolyte solution 65. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

66 [EASY] Electrolysis Principles & Cell Processes 66 — 9701/11/O/N/221/Q31**[1]**

An industrial electrolysis cell operates with electrolyte solution 66. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

67 [HARD] Electrolysis Principles & Cell Processes 67 — 9701/12/O/N/222/Q32**[1]**

An industrial electrolysis cell operates with electrolyte solution 67. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

68 [EASY] Electrolysis Principles & Cell Processes 68 — 9701/13/O/N/223/Q33**[1]**

An industrial electrolysis cell operates with electrolyte solution 68. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

69 [HARD] Electrolysis Principles & Cell Processes 69 — 9701/11/O/N/224/Q34**[1]**

An industrial electrolysis cell operates with electrolyte solution 69. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

70 [EASY] Electrolysis Principles & Cell Processes 70 — 9701/12/O/N/220/Q35**[1]**

An industrial electrolysis cell operates with electrolyte solution 70. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

71 [HARD] Electrolysis Principles & Cell Processes 71 — 9701/13/O/N/221/Q36**[1]**

An industrial electrolysis cell operates with electrolyte solution 71. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

72 [EASY] Electrolysis Principles & Cell Processes 72 — 9701/11/O/N/222/Q37**[1]**

An industrial electrolysis cell operates with electrolyte solution 72. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

73 [HARD] Electrolysis Principles & Cell Processes 73 — 9701/12/O/N/223/Q38**[1]**

An industrial electrolysis cell operates with electrolyte solution 73. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

74 [EASY] Electrolysis Principles & Cell Processes 74 — 9701/13/O/N/224/Q39**[1]**

An industrial electrolysis cell operates with electrolyte solution 74. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

75 [HARD] Electrolysis Principles & Cell Processes 75 — 9701/11/O/N/220/Q40**[1]**

An industrial electrolysis cell operates with electrolyte solution 75. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

76 [EASY] Electrolysis Principles & Cell Processes 76 — 9701/12/O/N/221/Q41**[1]**

An industrial electrolysis cell operates with electrolyte solution 76. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

77 [HARD] Electrolysis Principles & Cell Processes 77 — 9701/13/O/N/222/Q42**[1]**

An industrial electrolysis cell operates with electrolyte solution 77. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

78 [EASY] Electrolysis Principles & Cell Processes 78 — 9701/11/O/N/223/Q43**[1]**

An industrial electrolysis cell operates with electrolyte solution 78. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

79 [HARD] Electrolysis Principles & Cell Processes 79 — 9701/12/O/N/224/Q44**[1]**

An industrial electrolysis cell operates with electrolyte solution 79. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

80 [EASY] Electrolysis Principles & Cell Processes 80 — 9701/13/O/N/220/Q45**[1]**

An industrial electrolysis cell operates with electrolyte solution 80. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

81 [HARD] Electrolysis Principles & Cell Processes 81 — 9701/11/O/N/221/Q46**[1]**

An industrial electrolysis cell operates with electrolyte solution 81. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

82 [EASY] Electrolysis Principles & Cell Processes 82 — 9701/12/O/N/222/Q47**[1]**

An industrial electrolysis cell operates with electrolyte solution 82. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

83 [HARD] Electrolysis Principles & Cell Processes 83 — 9701/13/O/N/223/Q48**[1]**

An industrial electrolysis cell operates with electrolyte solution 83. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

84 [EASY] Electrolysis Principles & Cell Processes 84 — 9701/11/O/N/224/Q49**[1]**

An industrial electrolysis cell operates with electrolyte solution 84. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

85 [HARD] Electrolysis Principles & Cell Processes 85 — 9701/12/O/N/220/Q50**[1]**

An industrial electrolysis cell operates with electrolyte solution 85. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

86 [EASY] Electrolysis Principles & Cell Processes 86 — 9701/13/O/N/221/Q51**[1]**

An industrial electrolysis cell operates with electrolyte solution 86. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

87 [HARD] Electrolysis Principles & Cell Processes 87 — 9701/11/O/N/222/Q52**[1]**

An industrial electrolysis cell operates with electrolyte solution 87. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

88 [EASY] Electrolysis Principles & Cell Processes 88 — 9701/12/O/N/223/Q53**[1]**

An industrial electrolysis cell operates with electrolyte solution 88. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

89 [HARD] Electrolysis Principles & Cell Processes 89 — 9701/13/O/N/224/Q54**[1]**

An industrial electrolysis cell operates with electrolyte solution 89. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

90 [EASY] Electrolysis Principles & Cell Processes 90 — 9701/11/O/N/220/Q55**[1]**

An industrial electrolysis cell operates with electrolyte solution 90. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

91 [HARD] Electrolysis Principles & Cell Processes 91 — 9701/12/O/N/221/Q56**[1]**

An industrial electrolysis cell operates with electrolyte solution 91. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

92 [EASY] Electrolysis Principles & Cell Processes 92 — 9701/13/O/N/222/Q57**[1]**

An industrial electrolysis cell operates with electrolyte solution 92. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

93 [HARD] Electrolysis Principles & Cell Processes 93 — 9701/11/O/N/223/Q58**[1]**

An industrial electrolysis cell operates with electrolyte solution 93. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

94 [EASY] Electrolysis Principles & Cell Processes 94 — 9701/12/O/N/224/Q59**[1]**

An industrial electrolysis cell operates with electrolyte solution 94. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

95 [HARD] Electrolysis Principles & Cell Processes 95 — 9701/13/O/N/220/Q60**[1]**

An industrial electrolysis cell operates with electrolyte solution 95. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

96 [EASY] Electrolysis Principles & Cell Processes 96 — 9701/11/O/N/221/Q61**[1]**

An industrial electrolysis cell operates with electrolyte solution 96. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

97 [HARD] Electrolysis Principles & Cell Processes 97 — 9701/12/O/N/222/Q62**[1]**

An industrial electrolysis cell operates with electrolyte solution 97. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

98 [EASY] Electrolysis Principles & Cell Processes 98 — 9701/13/O/N/223/Q63**[1]**

An industrial electrolysis cell operates with electrolyte solution 98. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [B] Anions migrate to the cathode to donate electrons to positive ions.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

99 [HARD] Electrolysis Principles & Cell Processes 99 — 9701/11/O/N/224/Q64**[1]**

An industrial electrolysis cell operates with electrolyte solution 99. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Cations migrate to the anode where they undergo reduction.
- [C] Electrons travel through the electrolyte solution by hopping between water molecules.
- [D] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.

100 [EASY] Electrolysis Principles & Cell Processes 100 — 9701/12/O/N/220/Q65**[1]**

An industrial electrolysis cell operates with electrolyte solution 100. Under continuous operation, oxidation occurs at the anode while reduction occurs at the cathode. Which fundamental principle governs charge transfer at the electrodes?

- [A] Anions migrate to the cathode to donate electrons to positive ions.
- [B] Electrons flow from the anode through the external electrical circuit to the cathode, where reduction occurs.
- [C] Cations migrate to the anode where they undergo reduction.
- [D] Electrons travel through the electrolyte solution by hopping between water molecules.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [EASY] [HIGH-FREQUENCY EXAM REPEAT] Disproportionation of Chlorine in Cold Dilute Aqueous Alkali — 9701/11/M/J/23/Q28**[1]**

What are the oxidation states of chlorine in the products of the reaction between chlorine gas and cold, dilute aqueous sodium hydroxide (15 °C)?



- [A] -1 in NaCl and +5 in NaClO₃
- [B] 0 in NaCl and +1 in NaClO
- [C] -1 in NaCl and +1 in NaClO
- [D] -1 in NaCl and +3 in NaClO₂

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Disproportionation of Chlorine in Hot Concentrated Aqueous Alkali — 9701/12/M/J/23/Q28**[1]**

When chlorine gas is bubbled into hot, concentrated aqueous sodium hydroxide (70 °C), chlorate(V) ions are formed:



What are the oxidation states of chlorine in the reactant and the two chlorine-containing products?

- [A] Reactant: 0 in Cl_2 ; Products: -1 in NaCl and +5 in NaClO_3
- [B] Reactant: 0 in Cl_2 ; Products: -1 in NaCl and +1 in NaClO
- [C] Reactant: -1 in Cl_2 ; Products: 0 in NaCl and +5 in NaClO_3
- [D] Reactant: 0 in Cl_2 ; Products: -1 in NaCl and +7 in NaClO_4

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Deducing Oxidation State in Vanadium Oxycations — 9701/13/M/J/23/Q28**[1]**

Vanadium forms four aqueous ions in different oxidation states:

1: VO_2^+ (yellow)

2: VO_2^+ (blue)

3: V^{3+} (green)

4: V^{2+} (violet)

What are the oxidation states of vanadium in VO_2^+ and VO_2^+ ?

- [A] VO_2^+ : +4; VO_2^+ : +5
- [B] VO_2^+ : +3; VO_2^+ : +2
- [C] VO_2^+ : +5; VO_2^+ : +2
- [D] VO_2^+ : +5; VO_2^+ : +4

104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Electrolysis of Brine: Discharge at Electrodes — 9701/11/O/N/23/Q30**[1]**

In the industrial chlor-alkali diaphragm cell, concentrated aqueous sodium chloride is electrolysed.

Which half-equations represent the reactions occurring at the cathode and anode?

- [A] Cathode: $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$; Anode: $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
- [B] Cathode: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$; Anode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
- [C] Cathode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$; Anode: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$
- [D] Cathode: $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$; Anode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Electrolysis of Aqueous Copper(II) Sulfate with Carbon vs Copper Electrodes — 9701/12/O/N/23/Q30 **[1]**

When aqueous copper(II) sulfate, $\text{CuSO}_4(\text{aq})$, is electrolysed using inert carbon (graphite) electrodes, what product is formed at the anode, and what happens to the color of the electrolyte?

- [A] Anode product: Oxygen gas, $\text{O}_2(\text{g})$; Color of solution: Blue color gradually fades.
- [B] Anode product: Copper, $\text{Cu}(\text{s})$; Color of solution: Remains intensely blue.
- [C] Anode product: Sulfur dioxide, $\text{SO}_2(\text{g})$; Color of solution: Turns green.
- [D] Anode product: Hydrogen gas, $\text{H}_2(\text{g})$; Color of solution: Turns colorless.

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Faraday's Law: Calculation of Electroplated Mass — 9701/13/O/N/23/Q30**[1]**

An electroplating bath operates with a current of 2.50 A for 3860 seconds to deposit silver from an aqueous solution containing Ag^+ ions. [Faraday constant $F = 96500 \text{ C mol}^{-1}$; $A_r = 107.9$]

What mass of silver is deposited on the cathode?

- [A] 5.40 g
- [B] 21.58 g
- [C] 10.79 g
- [D] 1.08 g

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Ratio of Gas Volumes Discharged in Water Electrolysis — 9701/11/F/M/24/Q23**[1]**

During the electrolysis of dilute aqueous sodium sulfate, $\text{Na}_2\text{SO}_4(\text{aq})$, using inert platinum electrodes, what gases are evolved at the cathode and anode, and in what volume ratio (Cathode : Anode)?

- [A] Cathode: $\text{O}_2(\text{g})$; Anode: $\text{H}_2(\text{g})$; Volume ratio = 1 : 2
- [B] Cathode: $\text{H}_2(\text{g})$; Anode: $\text{O}_2(\text{g})$; Volume ratio = 2 : 1
- [C] Cathode: $\text{H}_2(\text{g})$; Anode: $\text{SO}_2(\text{g})$; Volume ratio = 1 : 1
- [D] Cathode: $\text{Na}(\text{s})$; Anode: $\text{O}_2(\text{g})$; Volume ratio = 4 : 1

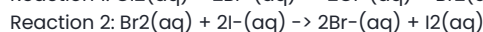
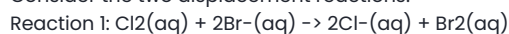
108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Balancing Redox Equation for Manganate(VII) Titration — 9701/12/F/M/24/Q23**[1]**

What is the stoichiometric ratio of MnO_4^- ions to $\text{C}_2\text{O}_4^{2-}$ (ethanedioate) ions in the fully balanced redox equation in acidic solution?
 $2\text{MnO}_4^-(\text{aq}) + 5\text{C}_2\text{O}_4^{2-}(\text{aq}) + 16\text{H}^+(\text{aq}) \rightarrow 2\text{Mn}^{2+}(\text{aq}) + 10\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\text{l})$

- [A] 1 : 5
- [B] 1 : 1
- [C] 2 : 3
- [D] 2 : 5

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Identifying the Strongest Reducing Agent from Redox Reactions — 9701/13/F/M/24/Q23**[1]**

Consider the two displacement reactions:



Which halogen species is the strongest reducing agent?

- [A] Iodide ion, I^-
- [B] Chloride ion, Cl^-
- [C] Chlorine, Cl_2
- [D] Bromide ion, Br^-

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Preferential Discharge of Halide Ions vs Hydroxide at the Anode — 9701/11/M/J/22/Q28**[1]**

Why are chloride ions discharged at the anode during the electrolysis of concentrated aqueous NaCl , even though standard electrode potential tables indicate that the oxidation of water (OH^-) is thermodynamically slightly easier?

- [A] Chloride ions carry a greater negative charge than hydroxide ions.
- [B] Oxygen gas cannot form bubbles on graphite electrodes.
- [C] The high concentration of Cl^- ions in concentrated brine shifts the electrode potential and kinetic overpotential, making chloride discharge kinetically favored over oxygen evolution.
- [D] Hydroxide ions are neutralized by sodium ions to form insoluble sodium oxide.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. Let the oxidation number of sulfur be x . The oxidation number of oxygen is -2 . The sum of oxidation numbers equals the net charge of the ion: $2x + 3(-2) = -2 \rightarrow 2x - 6 = -2 \rightarrow 2x = +4 \rightarrow x = +2$.	[1]
2	D	Option D is correct. In $\text{Cr}_2\text{O}_7^{2-}$, each oxygen has an oxidation state of -2 . Total charge = -2 . $2(\text{Cr}) + 7(-2) = -2 \rightarrow 2(\text{Cr}) - 14 = -2 \rightarrow 2(\text{Cr}) = +12 \rightarrow \text{Cr} = +6$.	[1]
3	A	Option A is correct. A disproportionation reaction is a redox reaction in which atoms of the same element in the same substance are simultaneously oxidized and reduced. In $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$, chlorine starts at oxidation state 0 in Cl_2 . In NaCl it is reduced to -1 , while in NaClO it is oxidized to $+1$. Reaction B is simple combination. Reaction C is single displacement. Reaction D is a non-redox precipitation.	[1]
4	C	Option C is correct. With cold dilute alkali, chlorine disproportionates into Cl^- (-1) and ClO^- ($+1$). With hot concentrated alkali (70°C), the chlorate(I) ion is unstable and disproportionates further according to: $3\text{Cl}_2 + 6\text{OH}^- \rightarrow 5\text{Cl}^- + \text{ClO}_3^- + 3\text{H}_2\text{O}$. The oxidation state of chlorine changes from 0 in Cl_2 to -1 in Cl^- and $+5$ in ClO_3^- .	[1]
5	A	Option A is correct. Each Cr atom changes oxidation state from $+6$ in $\text{Cr}_2\text{O}_7^{2-}$ to $+3$ in Cr^{3+} (a gain of 3 electrons per chromium atom). Since there are 2 chromium atoms, total electrons gained $x = 2 \times 3 = 6$ electrons. Checking charges: Left = $-2 + 14 - 6 = +6$; Right = $2(+3) = +6$ (balanced).	[1]
6	D	Option D is correct. Fluorine is the only element more electronegative than oxygen (Pauling value 3.98 vs 3.44). In OF_2 , each fluorine atom has an oxidation number of -1 . By charge neutrality: $x + 2(-1) = 0 \rightarrow x = +2$. In peroxides (H_2O_2 , BaO_2), oxygen is -1 . In superoxides (KO_2), oxygen is -0.5 .	[1]
7	B	Option B is correct. Assigning oxidation numbers: In $-\text{CH}_2\text{OH}$, C is bonded to H ($+1$), H ($+1$), O (-2), and C (0): $x + 2(+1) + (-2) + 0 = 0 \rightarrow x = -1$. In $-\text{CHO}$, C is bonded to H ($+1$), $=\text{O}$ (-2), and C (0): $x + 1 - 2 + 0 = 0 \rightarrow x = +1$. In $-\text{COOH}$, C is bonded to $=\text{O}$ (-2), $-\text{O}$ (-1), and C (0): $x - 2 - 1 + 0 = 0 \rightarrow x = +3$. Oxidation state increases by 2 units at each oxidation stage.	[1]
8	C	Option C is correct. A reducing agent donates electrons to another species and is itself oxidized in the process. Fe^{2+} loses one electron per ion to form Fe^{3+} (oxidation state increases from $+2$ to $+3$). MnO_4^- is the oxidizing agent (Mn is reduced from $+7$ to $+2$).	[1]
9	B	Option B is correct. Carbon monoxide (CO) is a neutral ligand with a net charge of 0. Since the entire complex $\text{Fe}(\text{CO})_5$ is uncharged and neutral, the oxidation state of the iron metal atom is exactly 0.	[1]
10	A	Option A is correct. In H_2O_2 , oxygen has an oxidation state of -1 . In elemental O_2 gas, oxygen has an oxidation state of 0. Because oxygen loses electrons and its oxidation number increases from -1 to 0, H_2O_2 acts as a reducing agent, reducing $\text{Mn}^{(+7)}$ in MnO_4^- to $\text{Mn}^{(+2)}$.	[1]
11	D	Option D is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
12	C	Option C is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
13	A	Option A is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
14	C	Option C is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
15	B	Option B is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
16	D	Option D is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
17	C	Option C is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
18	A	Option A is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
19	D	Option D is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
20	B	Option B is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
21	B	Option B is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
22	D	Option D is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]
23	A	Option A is correct. In VO_2^+ , oxygen is -2 ; charge is $+1$: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2 ; charge is $+2$: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from $+5$ to $+4$, corresponding to the addition of 1 electron.	[1]

23

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
51	D	Option D is correct. In VO_2^+ , oxygen is -2; charge is +1: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2; charge is +2: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from +5 to +4, corresponding to the addition of 1 electron.	[1]
52	C	Option C is correct. In VO_2^+ , oxygen is -2; charge is +1: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2; charge is +2: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from +5 to +4, corresponding to the addition of 1 electron.	[1]
53	A	Option A is correct. In VO_2^+ , oxygen is -2; charge is +1: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2; charge is +2: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from +5 to +4, corresponding to the addition of 1 electron.	[1]
54	C	Option C is correct. In VO_2^+ , oxygen is -2; charge is +1: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2; charge is +2: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from +5 to +4, corresponding to the addition of 1 electron.	[1]
55	B	Option B is correct. In VO_2^+ , oxygen is -2; charge is +1: $V + 2(-2) = +1 \rightarrow V = +5$. In VO_2^+ , oxygen is -2; charge is +2: $V - 2 = +2 \rightarrow V = +4$. The oxidation number of vanadium decreases from +5 to +4, corresponding to the addition of 1 electron.	[1]
56	D	Option D is correct. In molten NaCl , the only ions present are Na^+ and Cl^- (no water). At the negative cathode, Na^+ cations gain electrons (reduction): $\text{Na}^+ + e^- \rightarrow \text{Na(l)}$. At the positive anode, Cl^- anions lose electrons (oxidation): $2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2e^-$.	[1]
57	C	Option C is correct. In aqueous brine, ions present are Na^+ , H^+ , Cl^- , and OH^- . At the cathode, H^+ is more readily reduced than Na^+ due to its more positive reduction potential ($2\text{H}^+ + 2e^- \rightarrow \text{H}_2(\text{g})$). At the anode, in concentrated solution, Cl^- is preferentially discharged over OH^- ($2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2e^-$). As H^+ and Cl^- are removed, Na^+ and OH^- accumulate, leaving sodium hydroxide, NaOH(aq) , in solution.	[1]
58	A	Option A is correct. At the cathode: $2\text{H}^+ + 2e^- \rightarrow \text{H}_2(\text{g})$. At the anode: $2\text{H}_2\text{O} \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+ + 4e^-$. For every 4 moles of electrons passed through the circuit, 2 moles of H_2 gas are produced at the cathode and 1 mole of O_2 gas is produced at the anode. By Avogadro's hypothesis, gas volume is directly proportional to moles: Volume ratio of H_2 to $\text{O}_2 = 2 : 1$.	[1]
59	D	Option D is correct. Total electric charge $Q = I \times t = 5.00 \text{ A} \times 1930 \text{ s} = 9650 \text{ C}$. Moles of electrons passed = $Q / F = 9650 \text{ C} / 96500 \text{ C mol}^{-1} = 0.100 \text{ mol } e^-$. Cathode half-reaction: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu(s)}$. Moles of Cu deposited = moles of $e^- / 2 = 0.100 / 2 = 0.0500 \text{ mol}$. Mass of Cu = $0.0500 \text{ mol} \times 63.5 \text{ g mol}^{-1} = 3.175 \text{ g} \rightarrow 3.18 \text{ g}$.	[1]
60	B	Option B is correct. In copper refining: (1) Anode is made of thick slabs of impure copper (dissolves: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2e^-$), (2) Cathode is made of thin sheets of highly pure copper (deposits: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$), and (3) Electrolyte is aqueous acidified copper(II) sulfate.	[1]
61	B	Option B is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
62	D	Option D is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
63	A	Option A is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
64	C	Option C is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
65	A	Option A is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
66	D	Option D is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
67	B	Option B is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
68	C	Option C is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
69	B	Option B is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
70	A	Option A is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
71	D	Option D is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]

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Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
92	C	Option C is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
93	A	Option A is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
94	C	Option C is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
95	B	Option B is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
96	D	Option D is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
97	C	Option C is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
98	A	Option A is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
99	D	Option D is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
100	B	Option B is correct. In an electrolytic cell, oxidation (loss of electrons) occurs at the positive anode. These released electrons flow through the external metallic wiring to the negative cathode, where chemical reduction (gain of electrons by cations) takes place. Conduction within the electrolyte is strictly ionic, not electronic.	[1]
101	C	Option C is correct. In cold dilute sodium hydroxide (15 °C), chlorine (oxidation state 0) disproportionates according to: $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$. In sodium chloride (NaCl), chlorine is in the -1 oxidation state (reduction). In sodium chlorate(I) (NaClO), chlorine is in the +1 oxidation state (oxidation).	[1]
102	A	Option A is correct. In elemental chlorine gas, Cl_2 , the oxidation state is 0. In sodium chloride, NaCl, the oxidation state is -1 (gain of 1 electron). In sodium chlorate(V), NaClO_3 : $\text{Na}(+1) + \text{Cl}(x) + 3\text{O}(-2) = 0 \rightarrow 1 + x - 6 = 0 \rightarrow x = +5$ (loss of 5 electrons). This is a classic thermal disproportionation reaction.	[1]
103	D	Option D is correct. For the dioxovanadium(V) ion, VO_2^+ : $\text{V} + 2(-2) = +1 \rightarrow \text{V} - 4 = +1 \rightarrow \text{V} = +5$. For the oxovanadium(IV) ion, VO_2^{2+} : $\text{V} + 1(-2) = +2 \rightarrow \text{V} - 2 = +2 \rightarrow \text{V} = +4$.	[1]
104	B	Option B is correct. At the cathode, H^+ ions from water autoionisation are discharged in preference to Na^+ ions because H^+ has a more positive standard electrode potential ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$). At the anode, in concentrated solution, Cl^- ions are discharged in preference to OH^- ions ($2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$).	[1]
105	A	Option A is correct. At the cathode, Cu^{2+} ions are reduced to red-brown copper metal ($\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$). At the anode, with inert graphite, OH^- ions from water are oxidised in preference to sulfate ($4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$), evolving O_2 gas. Because Cu^{2+} ions are continuously removed from solution without replacement, the blue color of aqueous Cu^{2+} gradually fades to colorless, leaving dilute H_2SO_4 .	[1]
106	C	Option C is correct. Charge $Q = I \times t = 2.50 \text{ A} \times 3860 \text{ s} = 9650 \text{ C}$. Moles of electrons = $Q / F = 9650 \text{ C} / 96500 \text{ C mol}^{-1} = 0.100 \text{ mol e}^-$. Cathode reduction: $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$ (1 mole of e^- deposits 1 mole of Ag). Moles of Ag = 0.100 mol. Mass of Ag = $0.100 \text{ mol} \times 107.9 \text{ g mol}^{-1} = 10.79 \text{ g}$.	[1]
107	B	Option B is correct. In dilute Na_2SO_4 , water is electrolysed: Cathode: $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g})$; Anode: $4\text{OH}^- \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^-$. Overall cell reaction: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$. By Avogadro's law, 2 volumes of hydrogen gas are produced at the cathode for every 1 volume of oxygen gas produced at the anode (2 : 1 ratio).	[1]
108	D	Option D is correct. Oxidation: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$ (each ethanedioate loses 2 electrons). Reduction: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ (each manganate gains 5 electrons). To equalize electron transfer (10 electrons total): multiply oxidation half-equation by 5 and reduction half-equation by 2. Thus, 2 moles of MnO_4^- react with 5 moles of $\text{C}_2\text{O}_4^{2-}$ (ratio 2 : 5).	[1]
109	A	Option A is correct. A reducing agent donates electrons. Reaction 2 demonstrates that I^- reduces Br_2 to Br^- , while Reaction 1 demonstrates that Br^- reduces Cl_2 to Cl^- . The ability of halide ions to lose electrons (act as reducing agents) increases down Group 17: $\text{Cl}^- < \text{Br}^- < \text{I}^-$. Because iodide has the largest ionic radius, its valence electrons are held least tightly, making I^- the strongest reducing agent among the halogens.	[1]
110	C	Option C is correct. Although standard electrode potentials under 1.0 M standard conditions suggest OH^- discharge ($E^\circ = +0.40 \text{ V}$ or $+1.23 \text{ V}$) might compete with Cl^- discharge ($E^\circ = +1.36 \text{ V}$), the very high concentration of Cl^- ions in concentrated brine together with the high kinetic activation overpotential for oxygen evolution on graphite electrodes makes chloride discharge kinetically and practically dominant.	[1]